



3 Complex Analysis 2024
Tutorial 11
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Complex Logarithms

Problem 44 Calculate

$$(a) \log(-i), \quad (b) \log\left(\frac{1}{2}(1 + \sqrt{3}i)\right), \quad (c) i^{\frac{1}{2}}, \quad (d) i^{i+1}, \quad (e) (i+1)^i.$$

Solution.

$$(a) [\log(-1)] = \left\{-i\frac{\pi}{2} + i2\pi k : k \in \mathbb{Z}\right\},$$

$$(b) \left[\log\left(\frac{1}{2}(1 + \sqrt{3}i)\right)\right] = \left\{i\frac{\pi}{3} + i2\pi k : k \in \mathbb{Z}\right\},$$

$$(c) \left[i^{\frac{1}{2}}\right] = \left\{e^{\frac{1}{2}\left(i\frac{\pi}{2} + i2\pi k\right)} : k \in \mathbb{Z}\right\} = \left\{e^{i\pi\frac{1}{4}}, e^{i\pi\frac{5}{4}}\right\},$$

$$(d) [i^{i+1}] = \left\{e^{(i+1)i\left(\frac{\pi}{2} + 2\pi k\right)} : k \in \mathbb{Z}\right\} = \left\{e^{-\pi\left(\frac{1}{2} + 2k\right)} e^{i\frac{\pi}{2}} : k \in \mathbb{Z}\right\} = \left\{ie^{-\pi\left(\frac{1}{2} + 2k\right)} : k \in \mathbb{Z}\right\},$$

$$(e) [(i+1)^i] = \left\{e^{i(\log\sqrt{2} + i\left(\frac{\pi}{4} + 2k\right))} : k \in \mathbb{Z}\right\} = \left\{e^{-\pi\left(\frac{1}{4} + 2k\right)} e^{i\log\sqrt{2}} : k \in \mathbb{Z}\right\}.$$

Problem 45 Give an example that shows that, in general,

$$\text{Log}(zw) \neq \text{Log}(z) + \text{Log}(w) \quad (z, w \in \mathbb{C}).$$

Solution. For example, take $z = w = -1$, $zw = 1$. Then $\text{Log}(-1) = i\pi$, but $\text{Log}(1) = 0$.

Rouche's Theorem

Problem 46 For each of the following polynomials p , determine how many zeros p has in the unit disc $\{z \in \mathbb{C} : |z| < 1\}$:

$$(a) p(z) = z^7 - 5z^4 + z^2 - 2,$$

$$(b) p(z) = 2z^5 - z^3 + 3z^2 - z + 8,$$

$$(c) p(z) = z^9 - 2z^6 + z^2 - 8z - 2.$$

Solution.

(a) Take $f(z) = -5z^4$, $g(z) = z^7 + z^2 - 2$. For $|z| = 1$ we have

$$|f(z)| = 5, \quad |g(z)| \leq |z|^7 + |z|^2 + 2 = 4,$$

so that

$$|g(z)| < |f(z)|.$$

Since f has a zero of order 4 in the unit disc $\{z \in \mathbb{C} : |z| < 1\}$, $p = f + g$ has 4 zeros there.

(b) Take $f(z) = 8$ and $g(z) = 2z^5 - z^3 + 3z^2 - z + 8$, then for $|z| = 1$

$$|f(z)| = 8, \quad |g(z)| \leq 2|z|^5 + |z|^3 + 3|z|^2 + |z| = 7,$$

so that

$$|g(z)| < |f(z)|.$$

Since f has no zeros in the unit disc $\{z \in \mathbb{C} : |z| < 1\}$, $p = f + g$ has no zeros there.

(c) Take $f(z) = -8z$ and $g(z) = z^9 - 2z^6 + z^2 - 2$, then for $|z| = 1$

$$|f(z)| = 8, \quad |g(z)| \leq |z|^9 + 2|z|^6 + |z|^2 + 2 = 6,$$

so that

$$|g(z)| < |f(z)|.$$

Since f has one zero in the unit disc $\{z \in \mathbb{C} : |z| < 1\}$, $p = f + g$ has one zero there (notice that this zero must be real).

Problem 47 How many zeros does the polynomial $p(z) = z^4 - 5z + 1$ has in the annulus $\{z \in \mathbb{C} : 1 < |z| < 2\}$?

Solution.

First we determine the number of zeros of p in the disc $\{z \in \mathbb{C} : |z| < 2\}$. Take $f(z) = z^4$ and $g(z) = -5z + 1$. Then for $|z| = 2$

$$|f(z)| = |z|^4 = 16, \quad |g(z)| \leq 5|z| + 1 = 11,$$

so that

$$|g(z)| < |f(z)|, \quad |z| = 2.$$

Since f has 4 zeros in the unit disc $\{z \in \mathbb{C} : |z| < 2\}$ ($z = 0$ is a zero of order 4), $p = f + g$ has 4 zeros there.

Now let us determine the number of zeros of p in the closed disc $\{z \in \mathbb{C} : |z| \leq 1\}$.

1st solution: Take $f(z) = -5z$ and $g(z) = z^4 + 1$. For $|z| = 1$ we have

$$|f(z)| = 5|z| = 5, \quad |g(z)| \leq |z|^4 + 1 = 2,$$

so that

$$|g(z)| < |f(z)|, \quad |z| = 1. \quad (*)$$

Since f has one zero in the open unit disc $\{z \in \mathbb{C} : |z| < 1\}$, $p = f + g$ has one zero there.

Inequality $(*)$ also implies that $p = f + g$ does not have zeros with $|z| = 1$. Indeed, if $p = f + g = 0$, then $f = -g$ and $|f| = |g|$, which is not possible if $|z| = 1$ due to $(*)$. We conclude that p has one zero in the closed unit disc $\{z \in \mathbb{C} : |z| \leq 1\}$.

2nd solution: Now let us consider the disc $\{z \in \mathbb{C} : |z| < r\}$, where $r \in [1, 1 + \delta)$ with some small $\delta > 0$. Take $f(z) = -5z$ and $g(z) = z^4 + 1$. For $|z| = r$ we have

$$|f(z)| = 5r, \quad |g(z)| \leq |z|^4 + 1 = r^4 + 1,$$

and since $5 \cdot 1 < 1^4 + 1$ and the functions $r \mapsto 5r$, $r \mapsto r^4 + 1$ are continuous, there exists $\delta > 0$ such that $5r < r^4 + 1$ for all $r \in [1, 1 + \delta)$. Since f has one zero in the disc $\{z \in \mathbb{C} : |z| < r\}$, $p = f + g$ has one zero there. It follows that p has one zero in the closed unit disc $\{z \in \mathbb{C} : |z| \leq 1\}$.

We conclude that p has 3 zeros in the annulus $\{z \in \mathbb{C} : 1 < |z| < 2\}$.

Problem 48 Let f be a holomorphic function in the closed unit disc $\overline{D}(0; 1) = \{z \in \mathbb{C} : |z| \leq 1\}$. Assume that $|f(z)| < 1$ for all z with $|z| = 1$. How many solutions does the equation $f(z) = z$ has in the open unit disc $D(0; 1) = \{z \in \mathbb{C} : |z| < 1\}$?

Solution. We have that for all z with $|z| = 1$

$$|z| = 1 < |f(z)|.$$

This implies that z and $z - f(z)$ have the same number of zeros in the unit disc $\{z \in \mathbb{C} : |z| < 1\}$. Clearly the number of zeros is one.

Problem 49 (a) Show that for any $\lambda > 1$ the equation

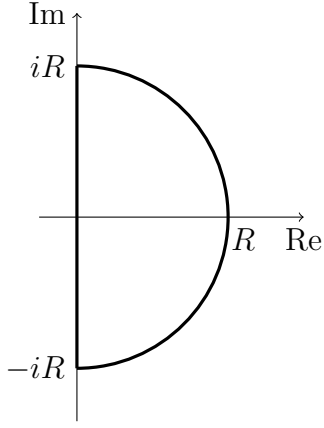
$$z + e^{-z} = \lambda$$

has exactly one solution in the right half-plane $\{z \in \mathbb{C} : \operatorname{Re} z > 0\}$.

(b) Prove that this solution is real.

Solution.

(a) Take a large $R > 0$ and consider the indicated region:



Take $f(z) = z - \lambda$ and $g(z) = e^{-z}$.

On the segment $[-iR, iR]$, z is purely imaginary and we have

$$|g(z)| = 1 < \lambda = |\operatorname{Re}(z - \lambda)| \leq |z - \lambda| = |f(z)|.$$

On the semicircle we have $z = Re^{it}$ with $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and

$$|g(z)| = \left|e^{-Re^{it}}\right| = \left|e^{-R(\cos t + i \sin t)}\right| = e^{-R \cos t} \leq 1 < \lambda.$$

If $R > 2\lambda$, then

$$|f(z)| = |z - \lambda| \geq |z| - \lambda = R - \lambda > \lambda.$$

We conclude that $|g(z)| < |f(z)|$ on the boundary of the region. Since $f(z) = z - \lambda$ has one zero inside the region, also $f(z) + g(z) = z - \lambda + e^{-z}$ has one zero, i.e. the equation $z + e^{-z} = \lambda$ has exactly one solution. This is true for any $R > 0$, and we conclude that the equation $z + e^{-z} = \lambda$ has exactly one solution in the right half-plane.

- (b) *1st solution:* We will show that if $z \in \mathbb{C}$ is such that $z + e^{-z} = \lambda$, then also $\bar{z} + e^{-\bar{z}} = \lambda$. Since the equation has only one solution, it follows that $z = \bar{z}$, i.e. z is real.

Let $z = x + iy$ with $x = \operatorname{Re} z$, $y = \operatorname{Im} z$. The equation $z + e^{-z} = \lambda$ turns into

$$x + iy + e^{-x}(\cos y - i \sin y) = \lambda,$$

and z is a solution if and only if $x + e^{-x} \cos y = \lambda$, $y - e^{-x} \sin y = 0$.

Now the equation $\bar{z} + e^{-\bar{z}} = \lambda$ is equivalent to

$$x - iy + e^{-x}(\cos y + i \sin y) = \lambda,$$

and leads to the same equations $x + e^{-x} \cos y = \lambda$, $y - e^{-x} \sin y = 0$.

We conclude that z is a solution if and only if $\bar{z}$ is a solution.

2nd solution: Consider the function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$, $\varphi(x) = x + e^{-x} - \lambda$. The function φ is continuous,

$$\varphi(0) = 1 - \lambda < 0, \quad \lim_{x \rightarrow +\infty} \varphi(x) = +\infty.$$

By the Intermediate Value Theorem, there exists $x_0 \in (0, \infty)$ with $\varphi(x_0) = 0$. This is the unique solution of $z + e^{-z} = \lambda$ in the right half-plane.